INCH-POUND

MIL-PRF-55681/1D
3 July 1997
SUPERSEDING

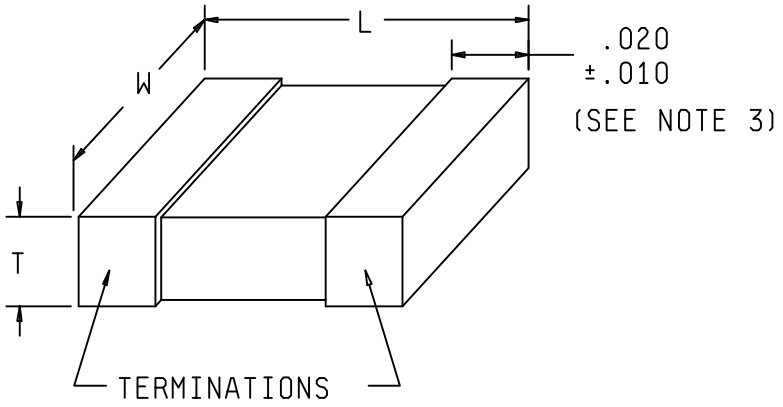
MIL-C-55681/1C
23 April 1992

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, CHIP, MULTIPLE LAYER, FIXED, CERAMIC DIELECTRIC, ESTABLISHED RELIABILITY AND NON-ESTABLISHED RELIABILITY, STYLES CDRO1, CDRO2, CDRO3, AND CDRO4

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55681.



Inches	mm
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.030	0.76

Dimensions				
Style	L ± .015 (0.38)	W ± .015 (0.38)	T	
			Min	Max
CDR01	.080 (2.03)	.050 (1.27)	.022 (0.56)	.055 (1.40)
CDR02	.180 (4.57)	.050 (1.27)	.022 (0.56)	.055 (1.40)
CDR03	.180 (4.57)	.080 (2.03)	.022 (0.56)	.080 (2.03)
CDR04	.180 (4.57)	.125 (3.18)	.022 (0.56)	.080 (2.03)

FIGURE 1. Styles CDR01, CDR02, CDR03, and CDR04 capacitors

DISTRIBUTION SATEMENT A. Approved for public release; distribution is unlimited.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. For style CDR01, termination width shall be .010 inch (0.25 mm), minimum; uncovered band width shall be .030 inch (0.76 mm), minimum.
4. Dimensions and tolerances are for bare chips. For solder coated terminations, for style CDR01 add .020 inch (0.51 mm) to the positive length tolerance and .015 inch (0.38 mm) to the positive width and thickness tolerance. For solder coated terminations for styles CDR02, CDR03, and CDR04, add .025 inch (0.64 mm) to the positive length tolerance and .015 inch (0.38 mm) to the positive width and thickness tolerance.

FIGURE 1. Styles CDR01, CDR02, CDR03, and CDR04 capacitors - Continued.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Capacitance value: See table I.

Capacitance tolerance: J = ±5 percent; K = ±10 percent; M = ±20 percent. See table I.

Rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Product level: Non-established Reliability - C; Established Reliability - M, P, R, and S.

Marking: In accordance with MIL-PRF-55681.

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TABLE I. CDR01, CDR02, CDR03, and CDR04 capacitors characteristics

Part or Identifying Number (PIN) 1/	Capacitance pF	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage V dc
CDRO1BP100B - - -	10	J, K	BP	100
CDRO1BP120BJ - -	12	J	BP	100
CDRO1BP150B - - -	15	J, K	BP	100
CDRO1BP180BJ - -	18	J	BP	100
CDRO1BP220B - - -	22	J, K	BP	100
CDRO1BP270BJ - -	27	J	BP	100
CDRO1BP330B - - -	33	J, K	BP	100
CDRO1BP390BJ - -	39	J	BP	100
CDRO1BP470B - - -	47	J, K	BP	100
CDRO1BP560BJ - -	56	J	BP	100
CDRO1BP680B - - -	68	J, K	BP	100
CDRO1BP820BJ - -	82	J	BP	100
CDRO1BP101B - - -	100	J, K	BP	100
CDRO1B-121B - - -	120	J, K	BP, BX	100
CDRO1B-151B - - -	150	J, K	BP, BX	100
CDRO1B-181B - - -	180	J, K	BP, BX	100
CDRO1BX221B - - -	220	K, M	BX	100
CDRO1BX271BK - -	270	K	BX	100
CDRO1BX331B - - -	330	K, M	BX	100
CDRO1BX391BK - -	390	K	BX	100
CDRO1BX471B - - -	470	K, M	BX	100
CDRO1BX561BK - -	560	K	BX	100
CDRO1BX681B - - -	680	K, M	BX	100
CDRO1BX821BK - -	820	K	BX	100
CDRO1BX102B - - -	1,000	K, M	BX	100
CDRO1BX122BK - -	1,200	K	BX	100
CDRO1BX152B - - -	1,500	K, M	BX	100
CDRO1BX182BK - -	1,800	K	BX	100
CDRO1BX222B - - -	2,200	K, M	BX	100
CDRO1BX272BK - -	2,700	K	BX	100
CDRO1BX332B - - -	3,300	K, M	BX	100
CDRO1BX392AK - -	3,900	K	BX	50
CDRO1BX472A - - -	4,700	K, M	BX	50
CDRO2BP221B - - -	220	J, K	BP	100
CDRO2BP271BJ - -	270	J	BP	100
CDRO2BX392BK - -	3,900	K	BX	100
CDRO2BX472B - - -	4,700	K, M	BX	100
CDRO2BX562BK - -	5,600	K	BX	100
CDRO2BX682B - - -	6,800	K, M	BX	100
CDRO2BX822BK - -	8,200	K	BX	100
CDRO2BX103B - - -	10,000	K, M	BX	100
CDRO2BX123AK - -	12,000	K	BX	50
CDRO2BX153A - - -	15,000	K, M	BX	50
CDRO2BX183AK - -	18,000	K	BX	50
CDRO2BX223A - - -	22,000	K, M	BX	50

1/ See footnote at end of table.

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TABLE I. CDR01, CDR02, CDR03, and CDR04 capacitors characteristics - Continued

PIN 1/	Capacitance pF	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage V dc
CDRO3BP331B - - -	330	J, K	BP	100
CDRO3BP391BJ - -	390	J	BP	100
CDRO3BP471B - - -	470	J, K	BP	100
CDRO3BP561BJ - -	560	J	BP	100
CDRO3BP681B - - -	680	J, K	BP	100
CDRO3BP821BJ - -	820	J	BP	100
CDRO3BP102B - - -	1,000	J, K	BP	100
CDRO3BX123BK - -	12,000	K	BX	100
CDRO3BX153B - - -	15,000	K, M	BX	100
CDRO3BX183BK - -	18,000	K	BX	100
CDRO3BX223B - - -	22,000	K, M	BX	100
CDRO3BX273BK - -	27,000	K	BX	100
CDRO3BX333B - - -	33,000	K, M	BX	100
CDRO3BX393AK - -	39,000	K	BX	50
CDRO3BX473A - - -	47,000	K, M	BX	50
CDRO3BX563AK - -	56,000	K	BX	50
CDRO3BX683A - - -	68,000	K, M	BX	50
CDRO4BP122BJ - -	1,200	J	BP	100
CDRO4BP152B - - -	1,500	J, K	BP	100
CDRO4BP182BJ - -	1,800	J	BP	100
CDRO4BP222B - - -	2,200	J, K	BP	100
CDRO4BP272BJ - -	2,700	J	BP	100
CDRO4BP332B - - -	3,300	J, K	BP	100
CDRO4BX393BK - -	39,000	K	BX	100
CDRO4BX473B - - -	47,000	K, M	BX	100
CDRO4BX563BK - -	56,000	K	BX	100
CDRO4BX823AK - -	82,000	K	BX	50
CDRO4BX104A - - -	100,000	K, M	BX	50
CDRO4BX124AK - -	120,000	K	BX	50
CDRO4BX154A - - -	150,000	K, M	BX	50
CDRO4BX184AK - -	180,000	K	BX	50

1/ Complete PIN shall include additional symbol(s) to indicate voltage-temperature limit (where applicable), capacitance tolerance (where applicable), termination finish, and product level.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, MC, OS, SH
Air Force - 19, 99
DLA - CC

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5910-1923-02)

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